

LOCALBRIEF

Private on-device lecture and meeting notes for Snapdragon-powered HP PCs

Project brief for the Snapdragon AI Lab Build & Present Challenge

Core idea

Turn a lecture or meeting recording into useful notes without requiring a cloud AI API.

1. The problem

Students and small teams often record lectures or meetings because it is hard to listen, take notes, and keep track of tasks at the same time. The obvious AI solution is to upload the audio to a cloud service. That creates a different problem: the recording may contain classroom discussions, client details, personal information, or unfinished ideas that the user did not intend to send to a third party.

LocalBrief is built around a simpler idea: for routine note-taking, the first pass can happen on the laptop itself.

2. The solution

LocalBrief takes an audio recording or transcript and produces a compact note pack: a short summary, key points, action items, a small glossary, and five revision questions. The product is intentionally narrow. It is not another general chatbot; it is a tool for turning spoken information into something the user can immediately study or act on.

3. What makes the Snapdragon version different

The Snapdragon build is designed around two local AI stages. For speech-to-text, the project uses the Qualcomm AI Hub Whisper Windows sample as the reference integration. Qualcomm describes this sample as on-device speech-to-text using Whisper through ONNX Runtime and the Snapdragon NPU. For the note-generation stage, the project uses the Qualcomm AI Hub Llama-v3.2-3B-Instruct model, which Qualcomm lists as supported on Snapdragon X Elite and Snapdragon X Plus 8-Core compute targets.

The application layer is separated from the model layer. That means the same product can be tested in demo mode, then connected to the Snapdragon model assets without redesigning the user experience. The final demonstration will show the actual runtime path and device measurements instead of estimated numbers.

4. User flow

1. Record or import a short lecture/meeting audio file.
2. Run local speech recognition and review the transcript.
3. Generate the note pack locally: summary, key points, actions, glossary and quiz questions.
4. Edit anything the model got wrong before saving.
5. Save or export the final notes from the same PC.

5. Technical architecture

Layer	Technology	Purpose
Interface	Streamlit / Python	Simple local desktop-style workflow in a browser tab.
Speech	Whisper + Qualcomm AI Hub Windows sample	Local transcription path designed to use the Snapdragon NPU.
Generation	Llama-v3.2-3B-Instruct	Turns the transcript into structured notes.
Runtime	ONNX Runtime GenAI; QNN/GenieX-compatible model assets	Hardware-aware on-device execution.
Storage	Local files / optional SQLite layer	Keep user content on the PC by default.
Evaluation	Benchmark script	Record latency, throughput and runtime path on the target PC.

6. Innovation in the product

- A useful pipeline instead of a generic chatbot: speech -> transcript -> structured note pack.
- Local-first workflow: the core experience does not require a cloud API.
- Two AI workloads in one application, making the Snapdragon NPU part of the product rather than a checkbox.
- A focused study mode: the same transcript can produce revision questions and a glossary, not only a summary.

7. Deployment and accessibility

The project is a small Python application with a local web UI. It can be launched from PowerShell and used in a normal browser. A demo mode keeps the interface testable before the target AI assets are installed. Snapdragon mode uses the local model assets prepared for the target machine. The UI uses simple text, clear sections, and short outputs so it works for users who are not AI specialists.

8. How the project will be evaluated

The final submission will include measurements collected on the actual Snapdragon-powered HP PC. The benchmark table will capture transcription time, LLM generation time, generated tokens per second, memory use, and the execution path used by the runtime. Quality will be checked on a small set of lecture/meeting samples by comparing whether the generated notes preserve the important points and do not invent actions or facts.

Metric	What will be shown in the final demo
Latency	Time from final input to usable output
Throughput	Generated tokens per second for the note-generation stage
Runtime path	CPU / GPU / NPU, with the intended Snapdragon path highlighted
Memory	Observed peak or steady-state memory during the model run
Quality	Human check of summary accuracy, action-item accuracy and useful questions

9. Scope and honest limitations

The repository does not pretend that every machine has the same model assets. The Snapdragon hardware path depends on the exact target processor, runtime package and model bundle. For that reason, the project includes a benchmark script and asks the final presenter to report only measurements actually captured on the target HP Snapdragon PC. This avoids presenting model-level Qualcomm reference numbers as if they were application benchmarks.

10. References used for the technical design

- **Qualcomm Windows on Snapdragon AI:** <https://www.qualcomm.com/developer/windows-on-snapdragon/windows-on-snapdragon-ai>
- **Qualcomm AI Hub - Compute Models:** <https://aihub.qualcomm.com/en-US/compute/models>
- **Qualcomm AI Hub - Llama-v3.2-3B-Instruct:**
https://aihub.qualcomm.com/compute/models/llama_v3_2_3b_instruct
- **Qualcomm AI Hub - Whisper Windows:**
https://aihub.qualcomm.com/apps/whisper_windows_py
- **Qualcomm AI Hub documentation:** <https://workbench.aihub.qualcomm.com/docs/>